

Assignment 20.3 Ai Assisted Coding

Htno:2303a51613

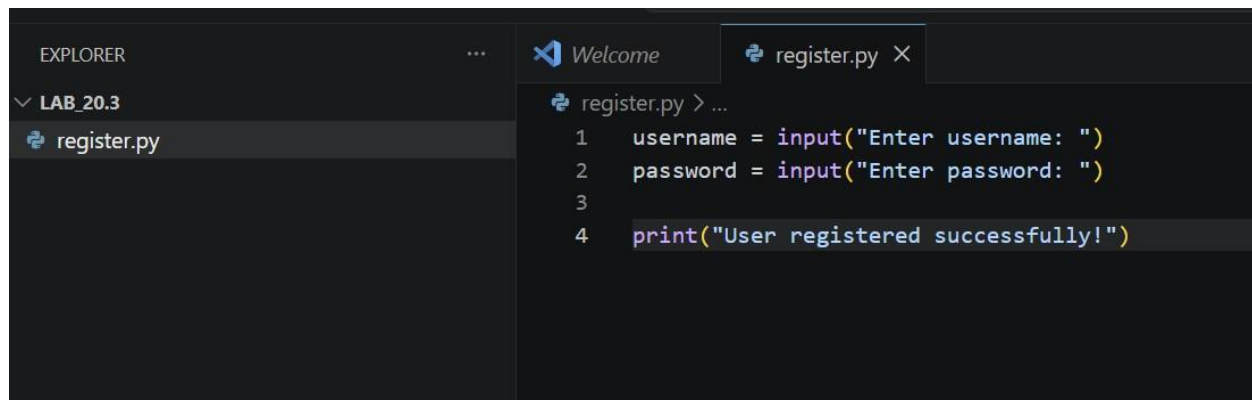
Btno:06

Task1: Password Strength & Validation Check

Prompt:Generate Python user registration program

Code:

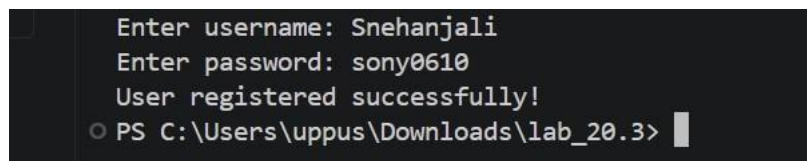
AI-Generated Code:



The screenshot shows the Visual Studio Code interface. On the left, the Explorer pane shows a folder named 'LAB_20.3' containing a file named 'register.py'. The main editor area shows the code for 'register.py' with the following lines:

```
1 username = input("Enter username: ")
2 password = input("Enter password: ")
3
4 print("User registered successfully!")
```

Output:

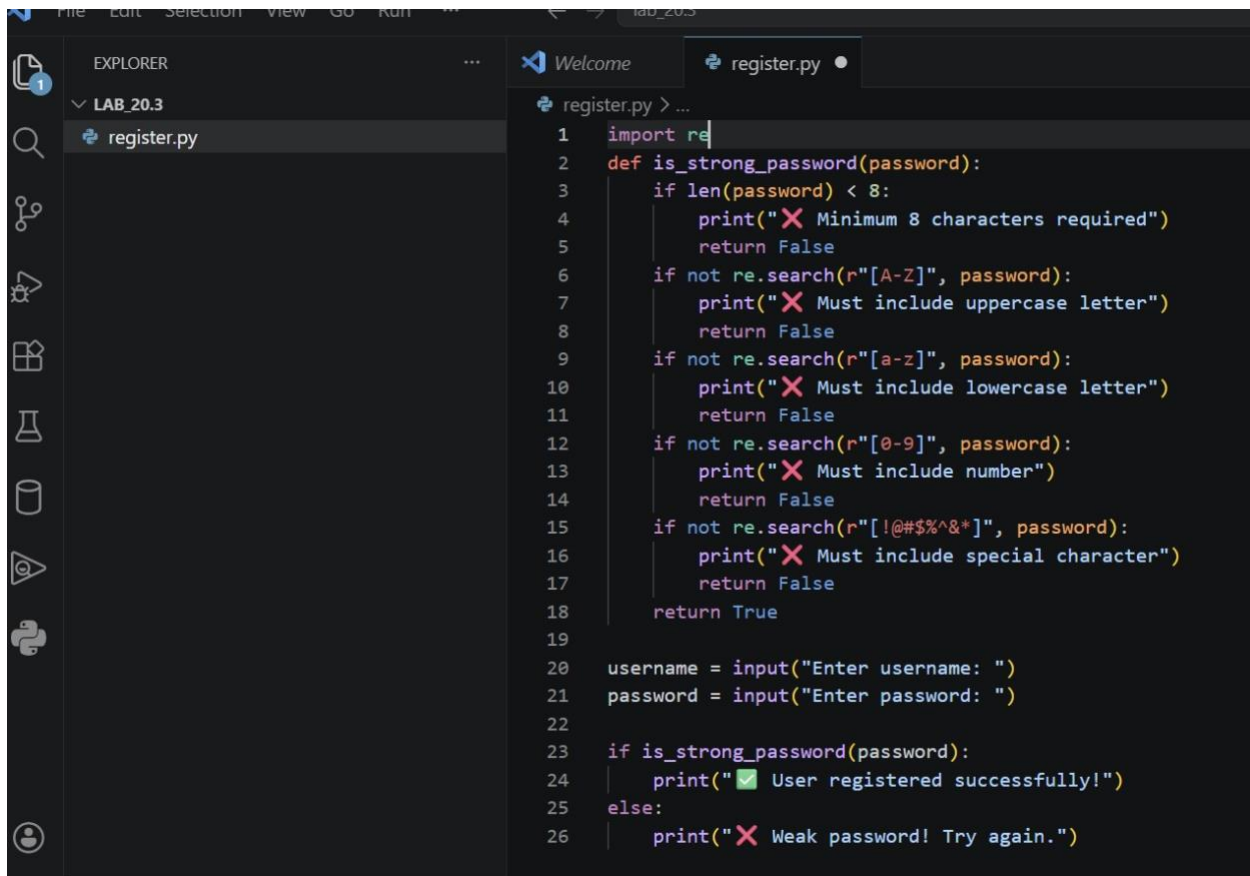


The screenshot shows the output of the program in a command prompt window. The output is as follows:

```
Enter username: Snehanjali
Enter password: sony0610
User registered successfully!
PS C:\Users\uppus\Downloads\lab_20.3>
```

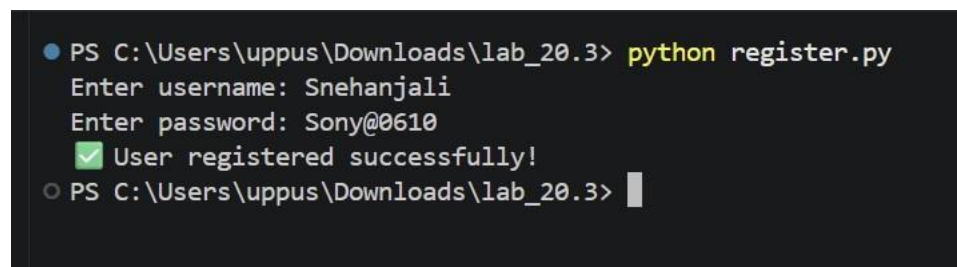
- The program accepts passwords without any validation checks.
- Weak passwords are allowed, making the system insecure.

Corrected Code:



```
1 import re
2 def is_strong_password(password):
3     if len(password) < 8:
4         print("❌ Minimum 8 characters required")
5         return False
6     if not re.search(r"[A-Z]", password):
7         print("❌ Must include uppercase letter")
8         return False
9     if not re.search(r"[a-z]", password):
10        print("❌ Must include lowercase letter")
11        return False
12    if not re.search(r"[0-9]", password):
13        print("❌ Must include number")
14        return False
15    if not re.search(r"[!@#$$%^&*]", password):
16        print("❌ Must include special character")
17        return False
18    return True
19
20 username = input("Enter username: ")
21 password = input("Enter password: ")
22
23 if is_strong_password(password):
24     print("✅ User registered successfully!")
25 else:
26     print("❌ Weak password! Try again.")
```

Output:



```
PS C:\Users\uppus\Downloads\lab_20.3> python register.py
Enter username: Snehanjali
Enter password: Sony@0610
✅ User registered successfully!
PS C:\Users\uppus\Downloads\lab_20.3>
```

Explanation:

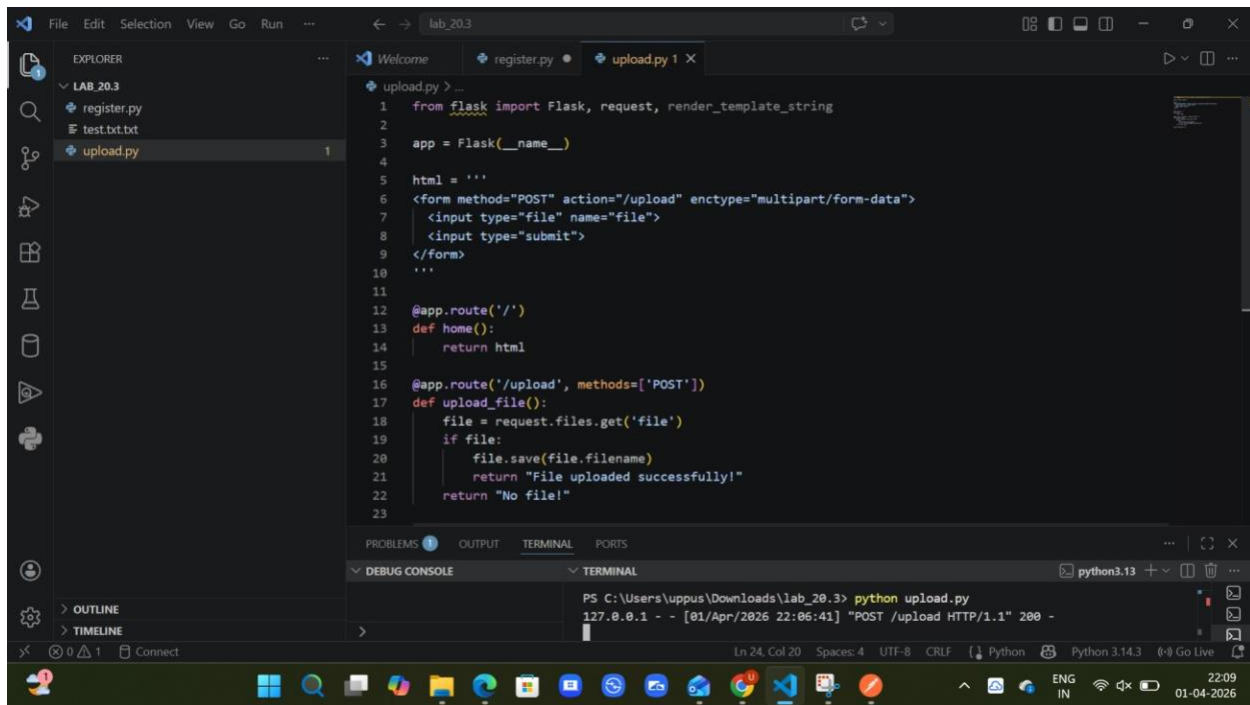
The initial program does not check the strength of the password. It accepts any password entered by the user, including weak ones. This makes the system insecure and vulnerable to attacks.

Task2: Insecure File Upload Handling

Prompt:

Generate a Flask-based file upload program that allows users to upload files.

Code: Ai Generated Code



```
1 from flask import Flask, request, render_template_string
2
3 app = Flask(__name__)
4
5 html = '''
6 <form method="POST" action="/upload" enctype="multipart/form-data">
7   <input type="file" name="file">
8   <input type="submit">
9 </form>
10 '''
11
12 @app.route('/')
13 def home():
14     return html
15
16 @app.route('/upload', methods=['POST'])
17 def upload_file():
18     file = request.files.get('file')
19     if file:
20         file.save(file.filename)
21         return "File uploaded successfully!"
22     return "No file!"
23
```

PS C:\Users\uppus\Downloads\lab_20.3> python upload.py
127.0.0.1 - - [01/Apr/2026 22:06:41] "POST /upload HTTP/1.1" 200 -

Output:

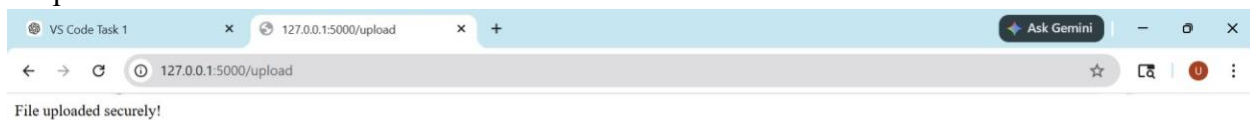


- The program allows uploading any type of file
- No validation of file extension

Corrected code:

```
1 from flask import Flask, request, render_template_string
2 import os
3
4 app = Flask(__name__)
5
6 UPLOAD_FOLDER = "uploads"
7 ALLOWED_EXTENSIONS = {"txt", "png", "jpg", "pdf"}
8
9 if not os.path.exists(UPLOAD_FOLDER):
10     os.makedirs(UPLOAD_FOLDER)
11
12 def allowed_file(filename):
13     return '.' in filename and filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS
14
15 html = '''
16 <form method="POST" action="/upload" enctype="multipart/form-data">
17   <input type="file" name="file">
18   <input type="submit">
19 </form>
20 '''
21
22 @app.route('/')
23 def index():
24     return render_template_string(html)
25
26 @app.route('/upload', methods=['POST'])
27 def upload():
28     if 'file' not in request.files:
29         return "No file part"
30     file = request.files['file']
31     if file.filename == '':
32         return "No selected file"
33     if file and allowed_file(file.filename):
34         filename = secure_filename(file.filename)
35         file.save(os.path.join(UPLOAD_FOLDER, filename))
36         return "File uploaded successfully!"
37     return "Invalid file type"
```

Output:



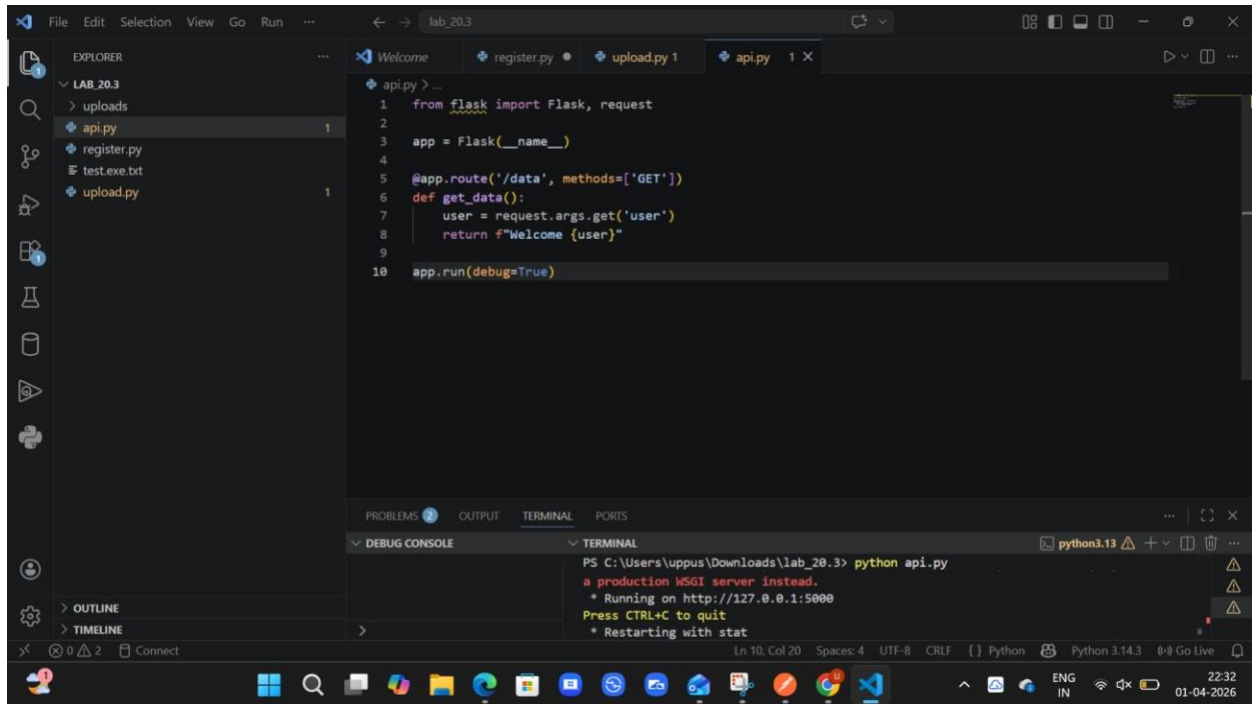
Explanation:

- The initial code allows uploading any file without checking its type, making it insecure.
- Malicious files like .exe can be uploaded, which may harm the system.

- The improved code restricts file types and allows only safe formats like .txt, .png, .jpg, and .pdf. Task3: API Security Review

Prompt:

Generate a Python Flask API endpoint for user data Code:

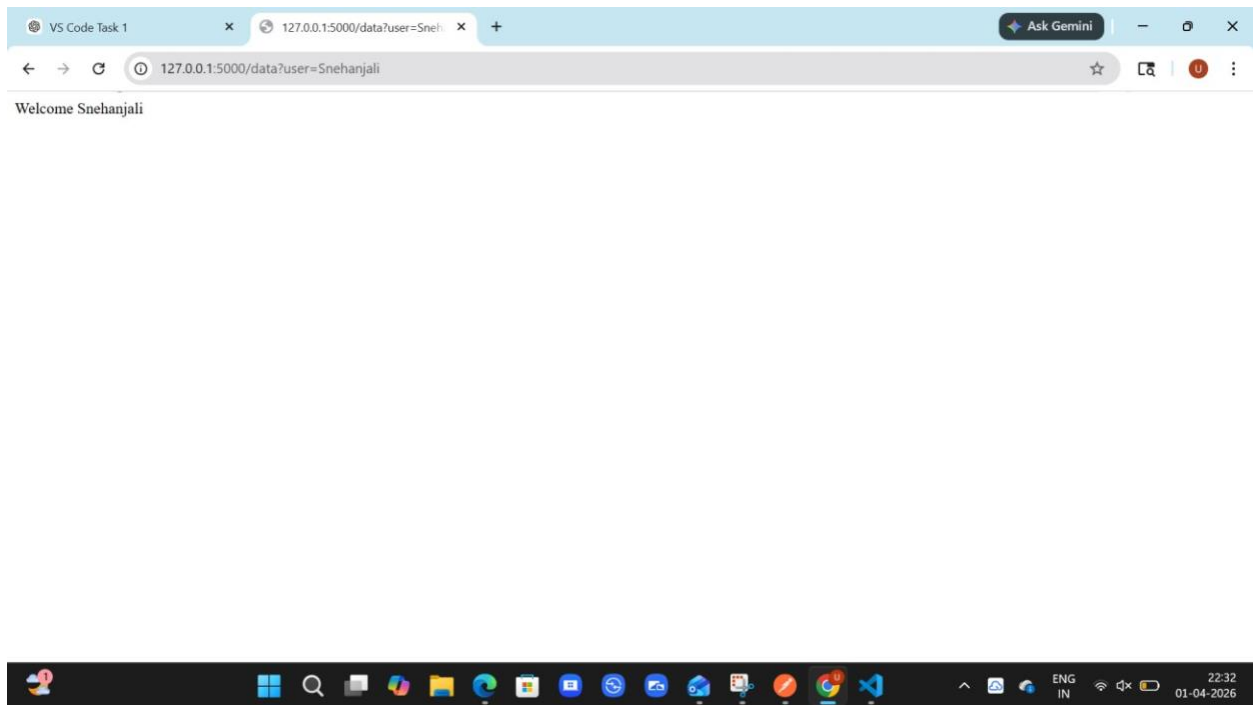


The screenshot shows a Visual Studio Code editor window with a project named 'lab_20.3'. The Explorer pane on the left shows a file structure with 'uploads' and 'api.py'. The main editor window displays the code for 'api.py', which is a Flask application. The code defines a route for '/data' that returns a welcome message for a user. The terminal at the bottom shows the command 'python api.py' being executed, and the output indicates that the server is running on http://127.0.0.1:5000.

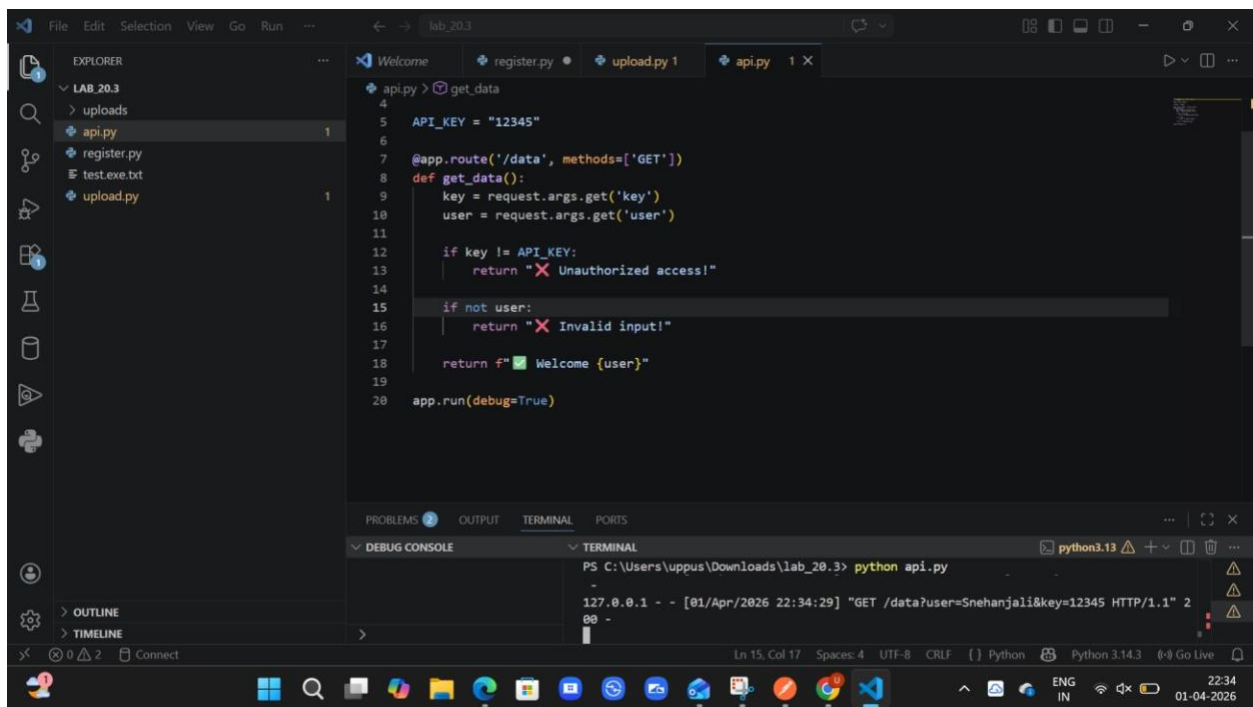
```
1 from flask import Flask, request
2
3 app = Flask(__name__)
4
5 @app.route('/data', methods=['GET'])
6 def get_data():
7     user = request.args.get('user')
8     return f"Welcome {user}"
9
10 app.run(debug=True)
```

PS C:\Users\uppus\Downloads\lab_20.3> python api.py
a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat

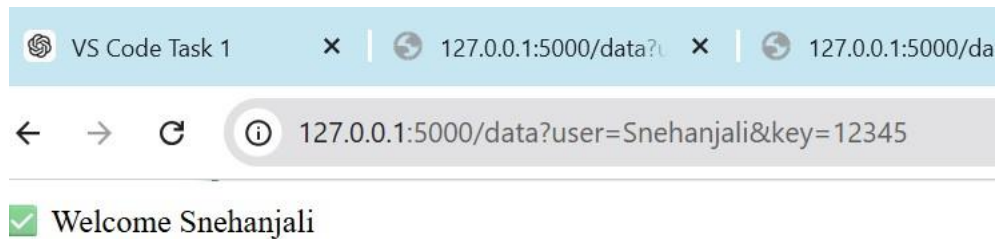
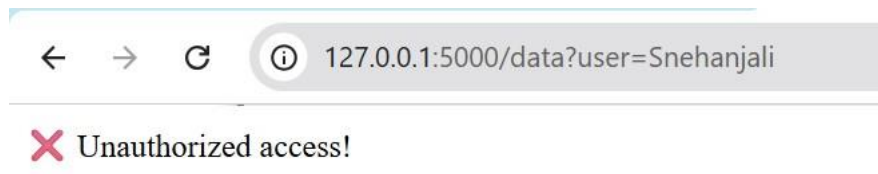
Output:



Corrected code:



Output:



Explanation:

- The initial API does not have authentication, allowing anyone to access it
- It does not validate user input, making it insecure
- The improved API uses an API key and input validation for secure access

Task 4: Cross-Site Scripting (XSS) Check

Prompt: Generate a feedback form using HTML and JavaScript

Code:

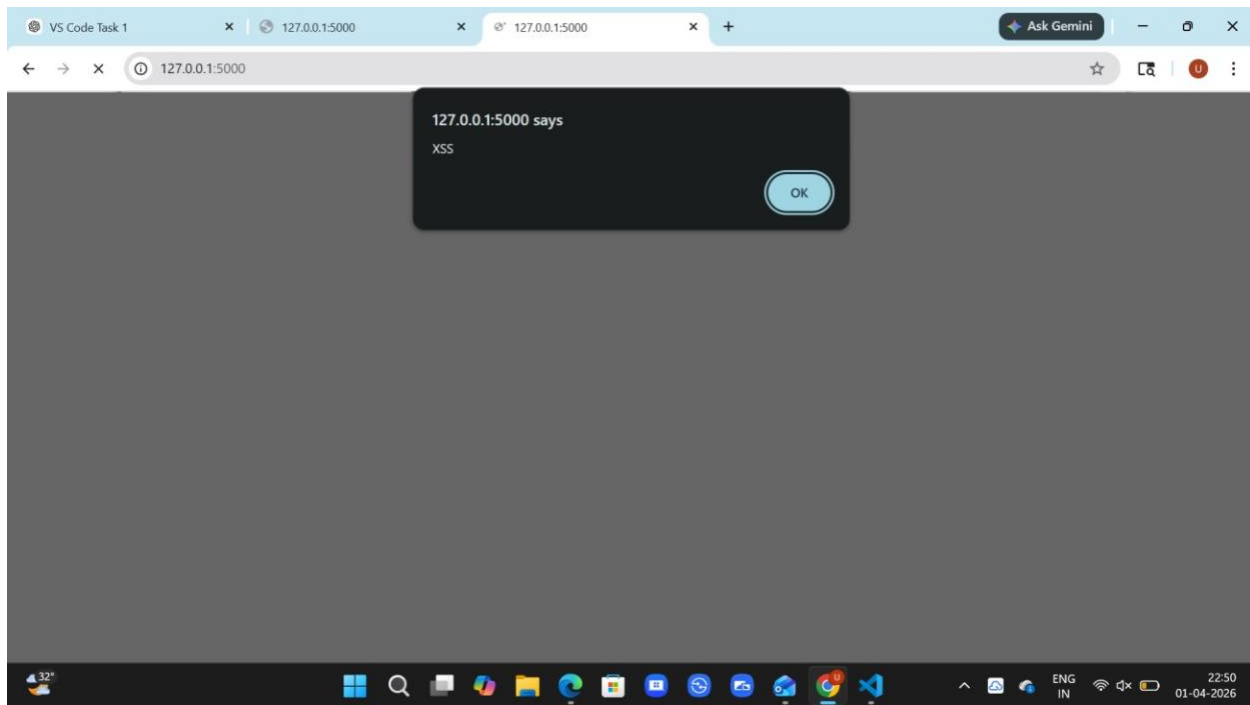
A screenshot of a Visual Studio Code editor window. The Explorer pane on the left shows a project named 'LAB_203' with files: `uploads`, `api.py`, `register.py`, `test.exe.txt`, `upload.py`, and `xss.py`. The main editor shows the content of `xss.py`:

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route('/', methods=['GET', 'POST'])
6 def form():
7     if request.method == 'POST':
8         name = request.form['name']
9         return f"Hello {name}"
10
11     return '''
12     <form method="POST">
13     <input type="text" name="name">
14     <input type="submit">
15     </form>
16     '''
17
18 app.run(debug=True)
```

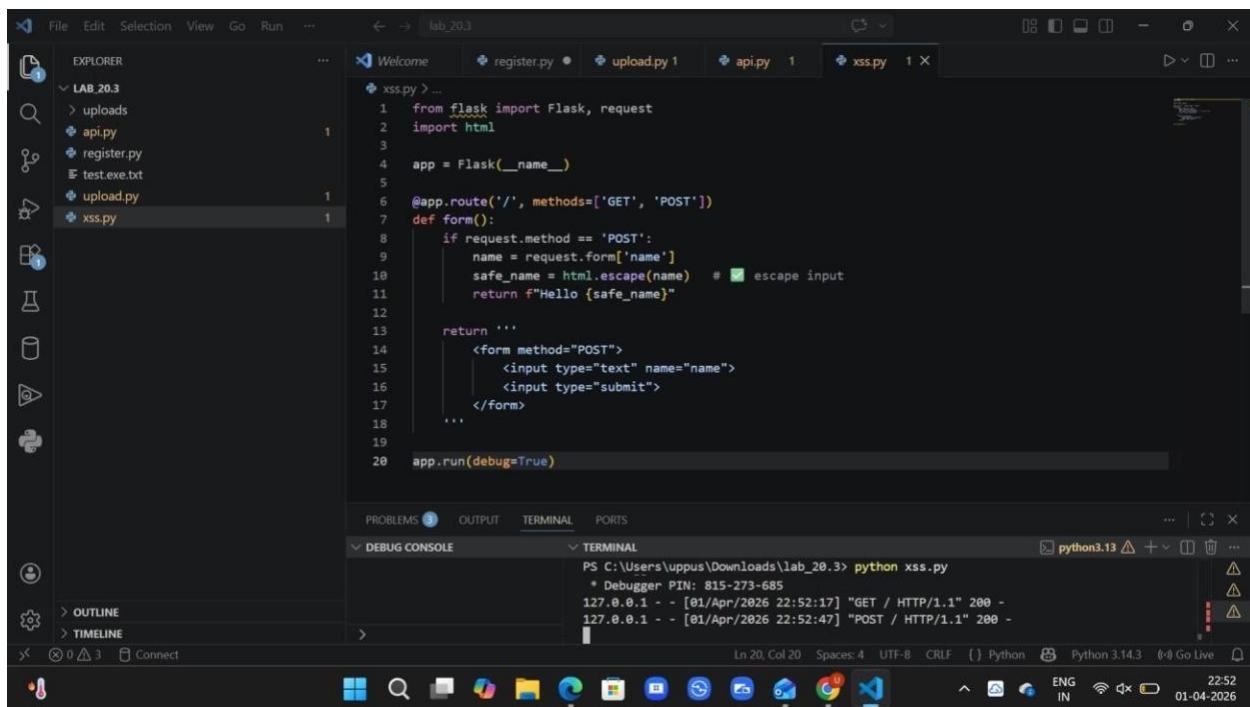
The bottom panel shows the TERMINAL with the following output:

```
PS C:\Users\uppus\Downloads\lab_20.3> python xss.py
127.0.0.1 - - [01/Apr/2026 22:47:28] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [01/Apr/2026 22:48:53] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [01/Apr/2026 22:49:08] "POST / HTTP/1.1" 200 -
```

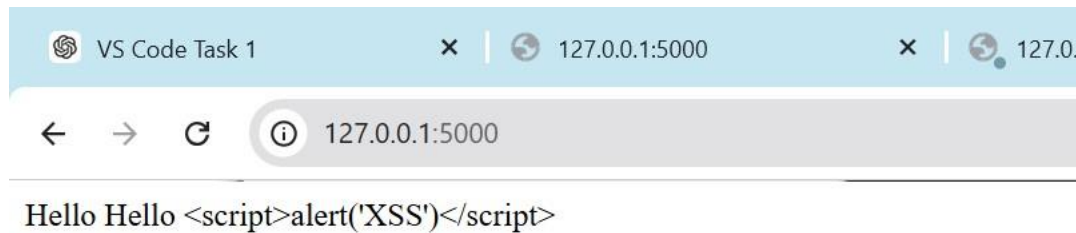
Output:



Corrected Code:



Output:



Explanation:

- The initial code displays user input directly without any validation, making it vulnerable to XSS attacks.
- Malicious scripts like `<script>alert('XSS')</script>` can execute in the browser.
- The secure code uses input escaping to prevent script execution and ensures safe output.

Task 5: SQL Injection Prevention

Prompt: Generate a Python script using SQLite to fetch user details from a database.

Code:

```
1 import sqlite3
2
3 conn = sqlite3.connect("users.db")
4 cursor = conn.cursor()
5
6 username = input("Enter username: ")
7
8 query = f"SELECT * FROM users WHERE username = '{username}'"
9 cursor.execute(query)
10
11 result = cursor.fetchall()
12 print(result)
```

Terminal Output:

```
[[ (1, 'Sneha'), (2, 'Admin') ]]
```

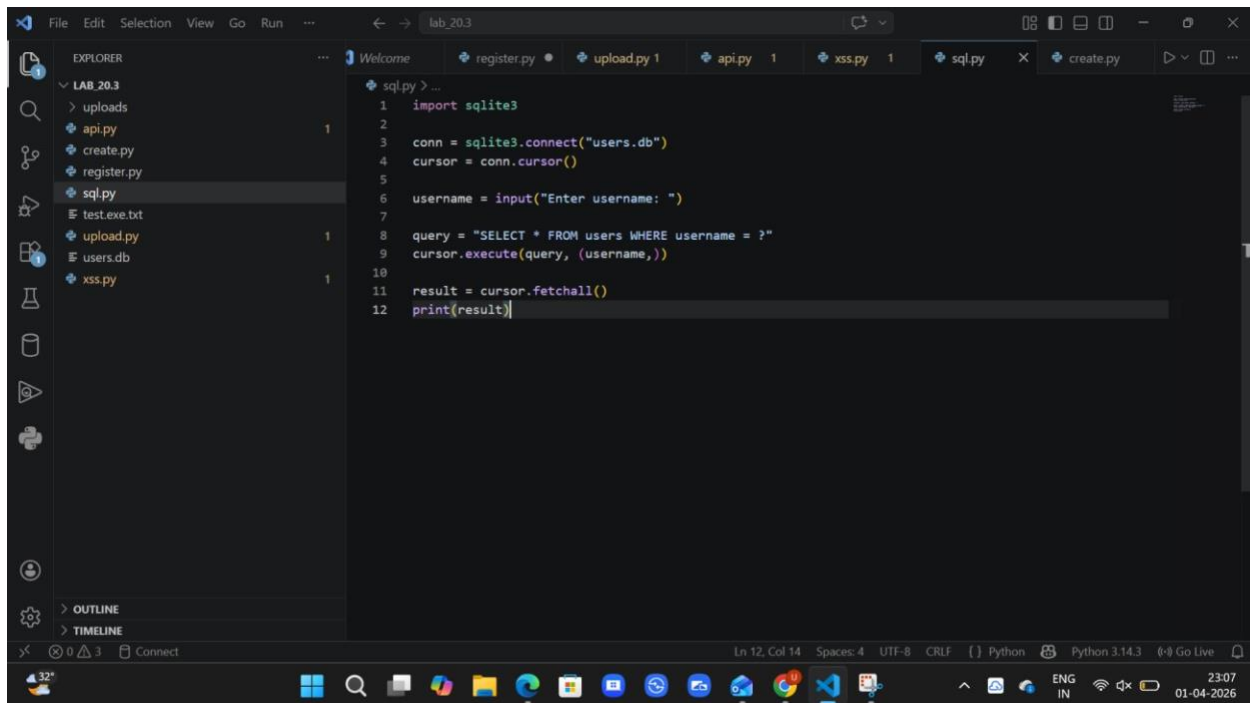
Output:

```

PS C:\Users\uppus\Downloads\lab_20.3> python sql.py
Enter username: ' OR '1'='1
[(1, 'Sneha'), (2, 'Admin')]
PS C:\Users\uppus\Downloads\lab_20.3>

```

Corrected code:



```

1 import sqlite3
2
3 conn = sqlite3.connect("users.db")
4 cursor = conn.cursor()
5
6 username = input("Enter username: ")
7
8 query = "SELECT * FROM users WHERE username = ?"
9 cursor.execute(query, (username,))
10
11 result = cursor.fetchall()
12 print(result)

```

Output:

```

PS C:\Users\uppus\Downloads\lab_20.3> python sql.py
Enter username: ' OR '1'='1
[]
PS C:\Users\uppus\Downloads\lab_20.3>

```

Explanation:

- The secure code uses parameterized queries (?) instead of string concatenation.
- User input is treated as data, not as part of the SQL query.
- This prevents SQL Injection and ensures safe database operations.

